

problem 970

Starting at zero, a kangaroo hops along the real number line in the positive direction. Each successive hop takes the kangaroo forward a uniformly random distance between 0 and 1. Let $H(n)$ be the expected number of hops needed for the kangaroo to pass n on the real line.

For example, $H(2) \approx 4.67077427047$. The first eight digits after the decimal point that are different from six are 70774270.

Similarly, $H(3) \approx 6.6665656395558899$. Here the first eight digits after the decimal point that are different from six are 55395558.

Find $H(10^6)$ and give as your answer the first eight digits after the decimal point that are different from six.

```
In [78]: def hops(n):
          d = 0
          number = 0
          while d < n:
              d += np.random.random(1)
              number += 1
          return(number)

          def H1(n):
              length = 1000000
              l = np.zeros(length)
              for i in range(length):
                  l[i] = hops(n)
              return(np.mean(l))
```

```
In [79]: H1(2)
```

```
Out[79]: 4.671758
```

```
In [80]: def H2(n):
          avg = hops(n)
          for i in range(1, 1000000):
              avg = ((i)*avg + hops(n))/(i+1)
          return(avg)
```

```
In [82]: H2(3)
```

```
Out[82]: 6.6639169999999959
```

```
In [ ]:
```